Assignment 1

For this week, complete the following problems from Epp:

Problem 6.1.4

Let $A = \{n \in \mathbb{Z} \mid n = 5r \text{ for some integer } r\}$ and $B = \{m \in \mathbb{Z} \mid m = 20s \text{ for some integer } s\}$. Prove or disprove each of the following statements.

a. $A \subseteq B$

b. $B \subseteq A$

Problem 6.1.8 (b–d)

Write in words how to read each of the following out loud. Then write each set using the symbols for union, intersection, set difference, or set complement.

b. $\{x \in U \mid x \in A \text{ or } x \in B\}$

c. $\{x \in U \mid x \in A \text{ and } x \notin B\}$

d. $\{x \in U \mid x \notin A\}$

Problem 6.2.17

Prove the following statement:
For all sets A, B, and C, if $A \subseteq B$ then $A \cup C \subseteq B \cup C$.

Problem 6.3.4

Find a counterexample to show that the following statement is false. Assume all sets are subsets of a universal set U.
For all sets A, B, and C, if $B \cup C \subseteq A$ then

$$(A - B) \cap (A - C) = \varnothing.$$

Problem 6.3.10

Prove the following statement:
For all sets A and B, if $A \subseteq B$ then $A \cap B^c = \varnothing$.

Additional Problem

Prove or disprove the following statement:
For all sets A and B, if $A \subseteq B$ then $B^c \subseteq A^c$.